

# Posterior Cruciate Ligament Injuries

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The treatment of posterior cruciate ligament (PCL) injuries is a controversial topic in orthopaedic surgery. In contrast to anterior cruciate ligament (ACL) injuries, for which an abundance of basic science and clinical data is available, the PCL has only recently become a topic of intense investigation. PCL injuries are less common compared with ACL injuries, and thus studies on outcomes are underpowered, making it difficult to draw definitive conclusions regarding management. However, recent biomechanical and clinical data have highlighted the importance of the PCL in knee stability and function. Injury to the PCL, which acts as the primary restraint to posterior tibial translation, may lead to instability, pain, diminished function, and eventually arthrosis.

The purpose of this chapter is to discuss the evaluation, diagnosis, and management of PCL injuries and to present the relevant historic and recent literature on these topics. After a brief review of the pertinent components of the history, physical examination, and imaging modalities, we discuss important considerations in decision-making and treatment options in patients with PCL injuries, as well as our preferred surgical technique and outcomes of surgical management of PCL injuries. Decision-making in this patient population is largely dependent on the grade of PCL injury and the presence or absence of concomitant ligamentous injuries in the knee. We also focus on the latest evidence regarding transtibial versus the tibial inlay technique, single- versus double-bundle methods of reconstruction, and the outcomes of these various surgical treatment options.

## HISTORY

The true incidence and prevalence of PCL injuries are unknown and difficult to estimate because many of these injuries, particularly prior to the introduction of magnetic resonance imaging (MRI), were not diagnosed.<sup>1</sup> The reported incidence of PCL injuries has differed depending on the population studied. The incidence is as low as 3% in the outpatient setting<sup>2</sup> and as high as 37% in the traumatic setting.<sup>3</sup> Traumatic injuries and sports-related injuries account for the majority of PCL injuries. A prospective analysis of patients presenting with acute hemarthrosis of the knee and diagnosed with a PCL injury demonstrated that 56.5% of patients were trauma victims, whereas 32.9% had a sports-related injury.<sup>3</sup> Yet isolated PCL injuries were infrequent in this cohort, with 96.5% being part of a multiligamentous

injury. Similarly, in a retrospective cohort of 494 patients with PCL insufficiency, Schulz et al.<sup>4</sup> found traffic accidents (45%) and athletic injuries (40%) to be the most common causes of injury. Among specific sports, the incidence of PCL injury tends to be greater in those involving contact, such as football, soccer, and rugby. In the cohort reviewed by Schulz and colleagues,<sup>4</sup> skiing and soccer were the sports with the highest incidence of PCL injuries. Overall, the incidence of PCL injury has been estimated to be relatively low in athletes across a variety of sports.<sup>5-8</sup>

Important information can be obtained from the history of the patient presenting with acute knee pain or trauma. Any patient with knee pain and swelling with a high-energy mechanism of injury should be suspected of having a PCL injury, another capsuloligamentous injury, or both. Patients commonly report the inability to bear weight, instability, and decreased knee range of motion (ROM). In contrast to ACL injuries, which often result from a noncontact event, PCL injuries are typically due to external trauma. The classic “dashboard injury” pattern results from a posteriorly directed force on the anterior aspect of the proximal tibia with the knee in a flexed position. In patients with a higher energy mechanism of injury, it is possible that a knee dislocation occurred at the time of the injury even if the knee is reduced at the time of the evaluation.

In athletics, the typical mechanism of isolated PCL injury is a direct blow to the anterior tibia (Fig. 100.1A) or a fall onto the knee with the foot plantar flexed. When the foot is in a position of dorsiflexion, the force is transmitted to the patella and distal femur, decreasing the risk of injury to the PCL (Fig. 100.2). Noncontact mechanisms of injury, although less common, have also been reported. Most commonly, this mechanism of injury occurs via forced hyperflexion of the knee (see Fig. 100.1B).<sup>9</sup> In a small cohort reported by Fowler and Messieh,<sup>9</sup> these injuries would often lead to incomplete tearing of the PCL with the posteromedial (PM) fibers intact. Knee hyperextension has also been described as a mechanism of injury, which is usually combined with a varus or valgus force that results in multiple ligament injury (see Fig. 100.1C). Isolated injuries may have more subtle presentations, with patients reporting stiffness, swelling, and pain located in the back of the knee or pain with deep knee flexion (squatting and kneeling). In contrast to acute ACL tears, a “pop” is not usually reported with acute isolated PCL injuries and athletes are often able to continue to play. Reports of anterior knee pain, difficulty ascending stairs, and instability

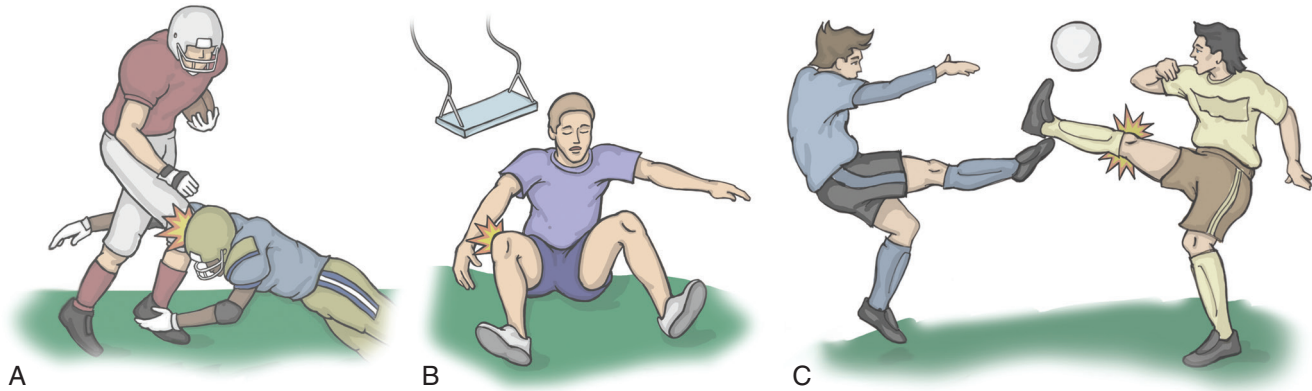
## Abstract

The treatment of posterior cruciate ligament (PCL) injuries is a controversial topic in orthopaedic surgery. In contrast to anterior cruciate ligament (ACL) injuries, for which an abundance of basic science and clinical data is available, the PCL has only recently become a topic of intense investigation. PCL injuries are less common compared with ACL injuries, and thus studies on outcomes are underpowered, making it difficult to draw definitive conclusions regarding management. However, recent biomechanical and clinical data have highlighted the importance of the PCL in knee stability and function. Injury to the PCL, which acts as the primary restraint to posterior tibial translation, may lead to instability, pain, diminished function, and eventually arthrosis.

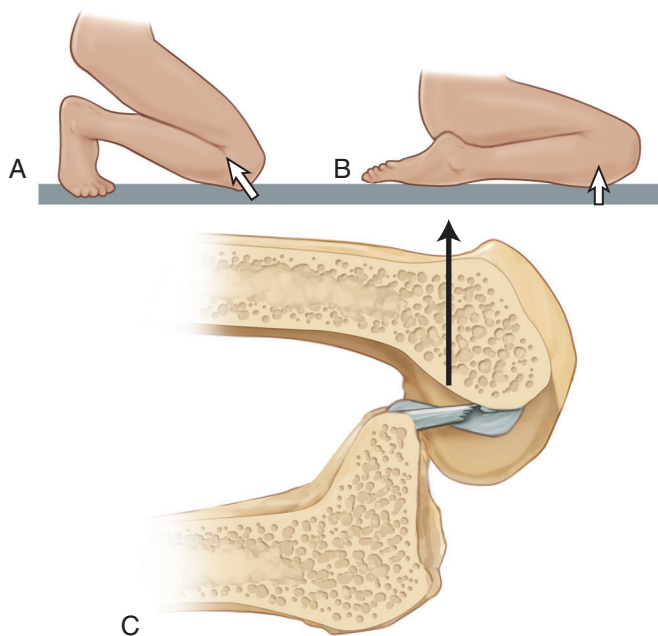
The purpose of this chapter is to discuss the evaluation, diagnosis, and management of PCL injuries and to present the relevant historic and recent literature on these topics. After a brief review of the pertinent components of the history, physical examination, and imaging modalities, we discuss important considerations in decision-making and treatment options in patients with PCL injuries, as well as our preferred surgical technique and outcomes of surgical management of PCL injuries. Decision-making in this patient population is largely dependent on the grade of PCL injury and the presence or absence of concomitant ligamentous injuries in the knee. We also focus on the latest evidence regarding transtibial versus the tibial inlay technique, single- versus double-bundle methods of reconstruction, and the outcomes of these various surgical treatment options.

## Keywords

posterior cruciate  
transtibial  
inlay  
single versus double bundle



**Fig. 100.1** Posterior cruciate ligament injuries are most frequently the result of a blow to the front of the flexed knee.



**Fig. 100.2** (A) Falling on a flexed knee with the foot in a dorsiflexed position spares injury to the posterior cruciate ligament (PCL) by transmitting the force to the patellofemoral joint. (B) Landing with the foot in a plantar flexed position injures the PCL because the posteriorly directed force is applied to the tibial tubercle. (C) Hyperflexion of the knee without a direct blow is a common mechanism of PCL injury in athletes.

are common when patients present in the chronic phase of an isolated PCL injury.<sup>10</sup>

## PHYSICAL EXAMINATION

### Posterior Drawer Test

The posterior drawer test was described initially by Hughston et al.<sup>11</sup> in 1976 and later by Clancy et al.<sup>12</sup> in 1983 and is considered the most accurate clinical test to assess the integrity of the PCL, with a sensitivity of 90% and 99% specificity.<sup>13,14</sup> The results of this examination also guide treatment recommendations. A posteriorly directed force is placed on the proximal tibia

with the patient lying supine and the knee flexed to 90 degrees. This test can be performed with the tibia in neutral, external, and internal rotation. It is important to remember that with a PCL injury, the tibia subluxes posteriorly. Thus it is important to first apply an anteriorly directed force to reduce the posterior subluxation before applying the posteriorly directed force (Fig. 100.3). In cases of isolated PCL tears, a decrease occurs in posterior tibial translation with internal tibial rotation.<sup>15</sup> The superficial medial collateral ligament (MCL) and posterior oblique ligament act as a secondary restraint with the tibia in internal rotation.<sup>16</sup> Translation is measured as the change in distance of step-off between the medial tibial plateau relative to the medial femoral condyle. It is critical to examine the contralateral knee, because the normal relationship between the medial tibial plateau and medial femoral condyle is variable, with the plateau normally resting on average 1 cm anterior to the condyle. Understanding this relationship is also critical in avoiding a false-positive anterior drawer test. The presence or lack of a firm end point should also be noted.

The amount of posterior translation observed during the posterior drawer test is used to grade the PCL injury. In grade I injuries, 0 to 5 mm of increased posterior translation is observed compared with the contralateral knee, but the anterior step-off of the plateau relative to the condyle is maintained. Grade II injuries are defined as those with 6 to 10 mm of posterior translation, which results in the plateau being flush with, but not posterior to, the medial femoral condyle. In both grade I and II injuries, the PCL is usually partially torn. With grade III injuries, posterior translation exceeds 10 mm and the medial tibial plateau displaces posterior to the medial femoral condyle during the posterior drawer test. This finding usually represents a complete tear of the PCL and could also represent a combined PCL and posterolateral corner (PLC) injury.

### Posterior Sag Test (Godfrey Test) and Quadriceps Active Test

The posterior sag test may be positive in patients with complete PCL tears or partial tears. The patient lies supine with the hip and knee flexed to 90 degrees and the limb supported at the foot by the examiner. The anterior aspect of the proximal tibia

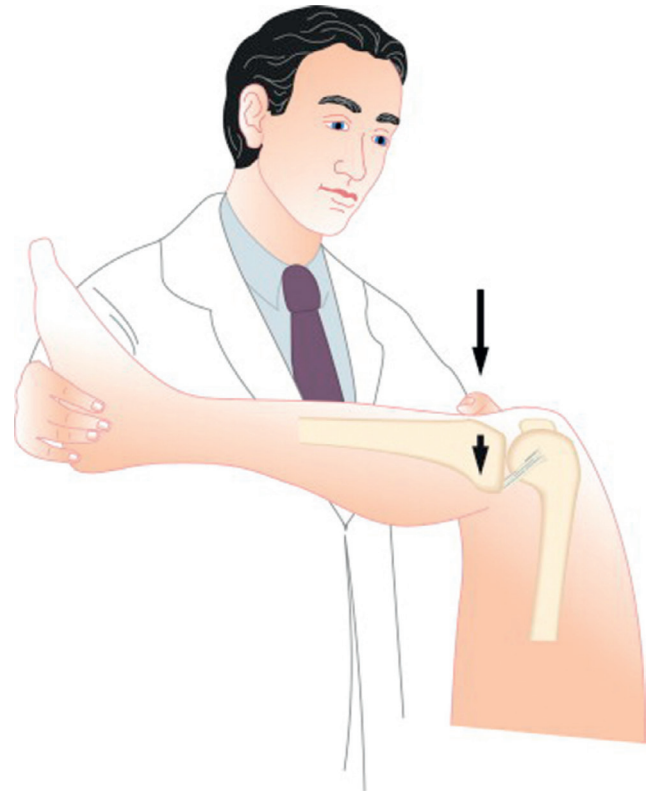


**Fig. 100.3** Assessing the tibial step-off before performing the posterior drawer examination. (Modified from Miller MD, Harner CD, Koshiwaguchi S. Acute posterior cruciate ligament injuries. In: Fu FH, Harner CD, Vince KG, eds. *Knee Surgery*. Vol. 1. Baltimore: Williams & Wilkins; 1994.)

is viewed from the side and compared with the uninjured, contralateral knee. Gravity displaces the tibia posterior to the femur in the case of a complete tear (Fig. 100.4). The quadriceps active test can aid in the diagnosis of complete tears. With this test, the patient lies supine and the knee is placed at 90 degrees of flexion. While the examiner stabilizes the foot, the patient is asked to contract the quadriceps isometrically. In the presence of a complete tear of the PCL (grade III), the patient will achieve dynamic reduction of the posteriorly displaced tibia.

### External Rotation of the Tibia (Dial Test)

The dial test is performed to evaluate for concomitant injuries to the PLC, which will affect decision-making and treatment options because these patients are more likely to require surgery. The dial test is performed with the patient positioned prone or supine, while an external rotation force is applied to both feet with the knee positioned at 30 degrees and then 90 degrees of flexion. The degree of external tibial rotation is measured by comparing the medial border of the foot with the axis of the femur. It is essential to compare the results with the contralateral



**Fig. 100.4** A positive Godfrey test. (Modified from Miller MD, Harner CD, Koshiwaguchi S. Acute posterior cruciate ligament injuries. In: Fu FH, Harner CD, Vince KG, eds. *Knee Surgery*. Vol. 1. Baltimore: Williams & Wilkins; 1994.)

side because wide variability of external rotation is possible at these positions.<sup>18,19</sup> More than a 10-degree side-to-side difference is considered abnormal.<sup>20</sup> At all degrees of knee flexion, the popliteus complex portion of the PLC is the primary restraint to external rotation, but its effect is maximal at 30 degrees. An increase of 10 degrees or more of external rotation at 30 degrees of knee flexion, but not at 90 degrees, is considered diagnostic of an isolated PLC injury.<sup>21</sup> Increased external rotation at both 30 and 90 degrees of knee flexion suggests a combined PCL and PLC injury.

### Reverse Pivot-Shift Test

The reverse pivot-shift test is also used to assess combined injuries and is performed with the patient supine. The knee is passively extended from 90 degrees of flexion with the foot externally rotated and a valgus force applied to the tibia. A positive result is observed when the posteriorly subluxed lateral tibial plateau is abruptly reduced by the iliotibial band at 20 to 30 degrees of flexion. A positive test typically indicates injury to the PCL and another capsuloligamentous structure, usually the PLC.<sup>22</sup>

### Collateral Ligament Assessment

Varus and valgus stress tests are used to assess the lateral collateral ligament (LCL) portion of the PLC. The tests are performed with the knee in full extension and in 30 degrees of flexion. Although an isolated PCL injury does not significantly affect varus or valgus stability, increased varus opening at 30 degrees



of knee flexion indicates an injury to the LCL and possibly the popliteus complex. If a significant degree of varus opening is noted at full extension, a combined injury of the PLC, PCL, and/or ACL is likely present.<sup>24,25</sup>

## Gait and Limb Alignment

The evaluation of gait and limb alignment is particularly important for persons with chronic injury of the PCL or the PLC. In these patients, varus alignment, external rotation, and varus thrust may be observed. Compromised function of the stabilizers of the lateral knee can lead to excessive posterolateral rotation and varus opening (or thrust) in the stance phase of gait.

## IMAGING

### Radiography

In the acute setting, plain radiographs of the knee should be performed, including bilateral standing anteroposterior, flexion posteroanterior 45 degrees with weight bearing, and Merchant patellar and lateral radiographs. These views are evaluated for posterior tibial subluxation, avulsion fractures, posterior tibial slope, and tibial plateau fractures. Tibial plateau fractures often indicate a high-energy injury with multiligament involvement. Bony avulsion fractures can be seen at the insertion of the PCL and at the fibular head, medial tibial plateau (medial Second fracture), or the tibial tubercle.<sup>26</sup> Identification of bony avulsion injuries of the PCL, when recognized acutely, may be repaired primarily with superior results compared with late reconstruction.<sup>26,27</sup> Identification of tibial tubercle fractures is also critical. The unopposed pull of the hamstrings causes posterior tibial subluxation in this scenario, which can become fixed within a short time, requiring open reduction. Medial Second fractures represent a medial capsular avulsion in PCL injuries that may be associated with a peripheral medial meniscus tear.<sup>28,29</sup> Lastly, long leg hip-to-ankle views are critical to evaluate overall lower extremity alignment, particularly varus, in chronic or revision cases.

Stress radiographs are not necessary to diagnose a PCL injury but may be helpful to differentiate between complete and partial PCL tears. However, these radiographs are most commonly used for research purposes. In a retrospective review of 21 patients with partial or complete PCL tears, Hewett and colleagues<sup>31</sup> found that stress radiographs were more accurate than KT-1000 measurements in diagnosing PCL tears. With the knee flexed to 70 degrees and an 89-N weight suspended from the tibia at the level of the tibial tubercle, a lateral radiograph was taken. The mean translation of the medial tibial plateau was 12.2 mm in the presence of a complete tear compared with 5.2 mm seen with a partial tear as confirmed with diagnostic arthroscopy. The magnitude of posterior tibial translation during stress radiography has been correlated with the presence of combined ligament injury. In a cadaveric study by Sekiya et al.,<sup>32</sup> the authors demonstrated that greater than 10 mm of posterior tibial translation on stress radiography correlated with the presence of a PLC injury in addition to a complete disruption of the PCL. It should be noted that the accuracy of stress radiography may be decreased by patient guarding and partial reduction of the tibia

with quadriceps activation; in addition, this infrequently performed examination is operator dependent. Stress radiographs can also be influenced by tibial rotation, and thus some authors have concluded that physical examination may be equally sensitive to stress radiographs in determining the presence and extent of a PCL tear.<sup>33</sup>

### Magnetic Resonance Imaging

MRI has become the imaging modality of choice for confirming the presence of an acute PCL tear and to diagnose associated injuries with a sensitivity of up to 100%.<sup>14,34,35</sup> The location and physical characteristics of the tear can also be assessed with MRI and may have implications for prognosis and treatment.<sup>36,37</sup> MRI may be less sensitive in the diagnoses of chronic tears. The normal PCL appears dark on T1- and T2-weighted sequences and is curvilinear in appearance.<sup>38</sup> In contrast, chronic tears of the PCL can heal and assume the aforementioned curvilinear appearance; thus MRIs are much less sensitive for chronic PCL tears, and the appearance of a normal shape of the ligament should not be used as a criterion for a normal PCL.<sup>39,40</sup>

Lastly, MRI provides important information on the status of the menisci, articular cartilage, and other ligaments in the knee, because concomitant injuries affect treatment decision-making and prognosis.<sup>41</sup> Bone bruises have been found in 83% of grade II and III PCL injuries on MRI, but in contrast to the bone bruise associated with ACL tears, the location is variable.<sup>42</sup> The utility of MRI for the diagnosis of associated injuries to the PLC has previously been evaluated. With use of thin-slice coronal oblique T1-weighted images through the entire fibular head, LaPrade and colleagues<sup>43</sup> were able to identify injury to the posterolateral structures with an accuracy of 68.8% to 94.4%, depending on the structure. Similarly, Theodorou et al.<sup>44</sup> found that MRI has an accuracy of 79% to 100% for the diagnoses of posterolateral injuries confirmed with arthroscopy.

### Bone Scan

Although a bone scan is not frequently used, it can be useful in the evaluation and management of chronic PCL injuries. In particular, patients with these injuries are predisposed to early medial and patellofemoral compartment chondrosis.<sup>12,45,46</sup> In the setting of an isolated PCL-deficient knee with medial or patellofemoral compartment pain and normal radiographs, a bone scan to assess these compartments may be indicated. Increased uptake suggests that surgical intervention may be beneficial,<sup>47</sup> although this supposition has not been proven definitively.

## DECISION-MAKING

Decision-making in the treatment of PCL injuries is dependent on the natural history of the disease, with most treatment recommendations made on the basis of symptoms, activity level, grade of the injury, and associated injuries. As with any orthopaedic ailment, operative intervention should be chosen only if it results in superior outcomes compared with nonsurgical management. Controversy exists regarding PCL treatment because the extent to which posterior laxity causes symptoms or accelerates the development of degenerative joint disease (DJD) is unclear.

However, a computational knee kinematic study by Kang et al. suggests that PCL deficiency leads to significantly increased contact forces on the patellofemoral joint as well as PLC structures during normal gait.<sup>48</sup> Furthermore, it is unknown whether reconstruction sufficiently mitigates laxity to result in clinical improvement and slow the development of DJD. Reducing posterior laxity with reconstruction may improve long-term outcomes in patients with PCL injuries, and yet residual laxity is common even after reconstruction.<sup>49,50</sup> Some investigators propose that isolated PCL tears follow a benign course in the short term without reconstruction<sup>1,9,47,51</sup> but that diminishing results may be seen at a later point.<sup>52</sup> To date, no study has demonstrated that PCL reconstruction can prevent the development of DJD.<sup>53</sup>

Controversy remains regarding indications for nonoperative versus operative management because few clinical studies have sufficient sample sizes and duration of follow-up to draw definitive conclusions. Additionally, a variety of PCL reconstructions are currently used, and the treatment of isolated PCL injuries is often reported in conjunction with combined injuries, such as PLC injuries, making outcome studies relatively heterogeneous. Currently most studies are retrospective in nature and use various outcome measurements, which make comparisons difficult. Until randomized prospective clinical trials are conducted, this debate will likely continue. The next section reviews the results of nonoperative management of PCL injuries and conclude the section with our decision-making rationale.

## Nonoperative Treatment

Many studies have found favorable outcomes when isolated PCL injuries are treated conservatively. Parolie and Bergfeld<sup>1</sup> evaluated patient satisfaction in 25 persons with isolated PCL tears that had resulted from sporting injuries at a minimum of 2 years of follow-up. These investigators found that 68% of patients returned to their previous level of activity and 80% were satisfied with their knee function. They evaluated laxity and found no correlation with DJD. More recently, Shelbourne and Muthukaruppan<sup>54</sup> prospectively evaluated 215 conservatively treated patients with isolated PCL tears. Their study focused on patients with grade II laxity or less. These investigators found that subjective scores did not correlate with the degree of laxity and mean scores did not decrease with time from injury. They were unable to identify any risk factors that would predict which patients would have a decline in knee function over time. Patel et al.,<sup>55</sup> in another recent retrospective review of 57 patients with grade A or B PCL tears, a grading system proposed by MacGillivray and colleagues<sup>56</sup> also found that functional scores did not correlate with the degree of PCL laxity. Lysholm knee scores were excellent in 40% and good in 52%. Patel et al.<sup>55</sup> found grade I medial compartment osteoarthritis (OA) in seven knees, grade II in three knees, and mild patellofemoral OA in four knees at an average of 6.8 years of follow-up. They concluded that most patients with acute, isolated PCL tears do well with nonoperative management at intermediate follow-up. Furthermore, a more recent study by Shelbourne and colleagues showed that at an average of 14.3 years follow-up 89% (39/44) of individuals with isolated grade I or II PCL injuries treated nonoperatively showed full ROM, good quadriceps strength, and minimal OA.<sup>57</sup>

Other investigators have also found good initial clinical outcomes with nonoperative treatment but have found deterioration at extended follow-up. Boynton and Tietjens<sup>52</sup> observed 38 patients with isolated tears for a mean of 13.4 years. Of these patients, eight had subsequent meniscal injuries and surgery. Of the remaining 30 patients with normal menisci, 24 (81%) had occasional pain, 17 (56%) had occasional swelling, and a positive increase in articular cartilage degeneration was seen on radiographs over time. Fowler and Messieh<sup>9</sup> prospectively followed up 13 patients with acute isolated PCL tears that were confirmed by arthroscopy and treated with physiotherapy. All patients had a good subjective functional score according to the Houston criteria, but objective scores were good in only 3 patients and only fair in the other 10 patients.

Although relatively good results have been observed with nonoperative treatment, it should be noted that many of the patients in these series had grade II laxity or less and not all patients achieved a normal outcome, especially patients with grade III injuries. The benign course observed may be due to the integrity of the secondary restraints and various portions of the PCL complex remaining intact in persons with less serious injuries. Tibial slope may also affect the stability of the PCL deficient knee. In a cadaveric study, increasing the posterior tibial slope decreased the static posterior instability of the PCL/PLC-deficient knee, whereas decreasing the tibial slope increased posterior instability and the magnitude of the reverse pivot-shift test.<sup>58</sup>

Despite acceptable clinical results with nonoperative treatment, it is well understood that PCL deficiency alters knee kinematics and the distribution of load during activity. In an uninjured knee, numerous biomechanical studies have shown that the PCL plays a significant role in tibiofemoral joint stability throughout an active and passive ROM especially in the presence of a posteriorly directed tibial force, while it has been shown that the PCL-deficient knee experiences increased contact pressures in the patellofemoral and medial compartments.<sup>48,59,60</sup> Logan et al.<sup>59</sup> evaluated the effect of PCL rupture on tibiofemoral motion during squatting with use of MRI. They concluded that PCL deficiency is similar to a medial meniscus resection and results in a “fixed” anterior subluxation of the medial femoral condyle (posterior subluxation of the medial tibial plateau). This subluxation changes the kinematics of the knee and may explain the increase in medial compartment OA seen in PCL-deficient knees. Currently more attention is being placed on additional injuries that are commonly associated with grade III tears that lead to greater instability and more severely altered biomechanics.

Although it is known that the kinematics of the knee are altered in the presence of a PCL injury, specific prognostic factors that predict outcome have proven elusive. In many studies the time from injury and objective instability have not correlated well with final outcome and radiographic changes. Surgical reconstruction is not recommended for isolated grade I injuries. Because many patients with isolated grade II posterior laxity only improve to grade I laxity with reconstruction, we agree with other authors that operative intervention in these patients may not offer improved outcome when compared with nonoperative treatment.<sup>49,54</sup> The treatment of acute isolated grade III PCL injuries is controversial. In these patients, some surgeons

favor a more aggressive approach involving PCL reconstruction, whereas others recommend nonsurgical treatment. In cases with greater than 10 mm of abnormal posterior laxity, the clinician should remember to have a high index of suspicion for a combined ligamentous injury involving the PLC.

Level I evidence does not currently exist to support strong recommendations on the management of PCL injuries. However, based on the previously described data, we recommend nonoperative management for the treatment of acute and chronic isolated grade I and II PCL injuries (Figs. 100.5 and 100.6). Operative management is reserved for chronic isolated grade III PCL injuries with symptoms of pain or instability when an adequate course of conservative treatment has failed. In addition, surgical treatment is usually recommended for acute and chronic combined ligamentous injuries. The treatment of acute grade III PCL tears is controversial, with some surgeons recommending PCL reconstruction and others recommending nonoperative treatment. Lastly, open reduction and internal fixation are recommended for acute avulsion fractures at the PCL tibial attachment site.

## TREATMENT OPTIONS

### Nonoperative Treatment

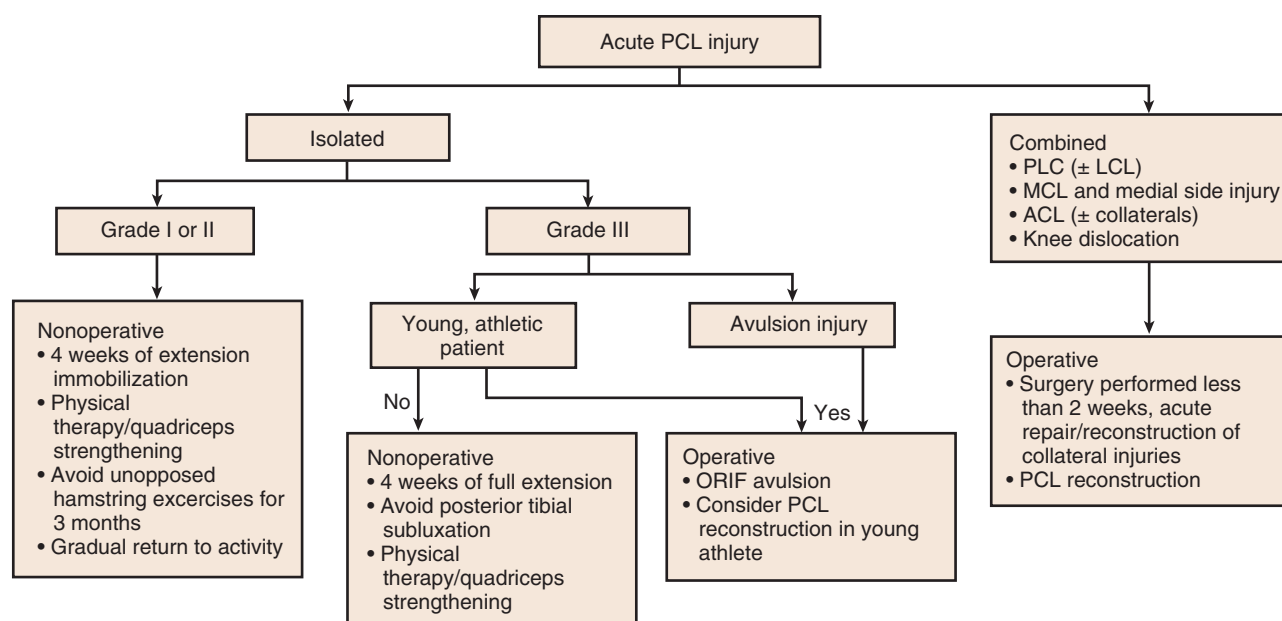
As discussed in the previous section, nonoperative treatment is recommended in patients with acute, isolated grade I or II PCL tears.<sup>1,45</sup> Nonoperative management is aimed at counteracting the forces of gravity and the hamstring muscles, which act to sublux the tibia posteriorly on the femur. Pierce and colleagues<sup>62</sup> have recently reviewed the literature on rehabilitation protocols for nonoperative and operative treatment of PCL injuries. Based on the finding of the reviewed studies, a three-phase rehabilitation

protocol was recommended. We follow a similar protocol at our institution.

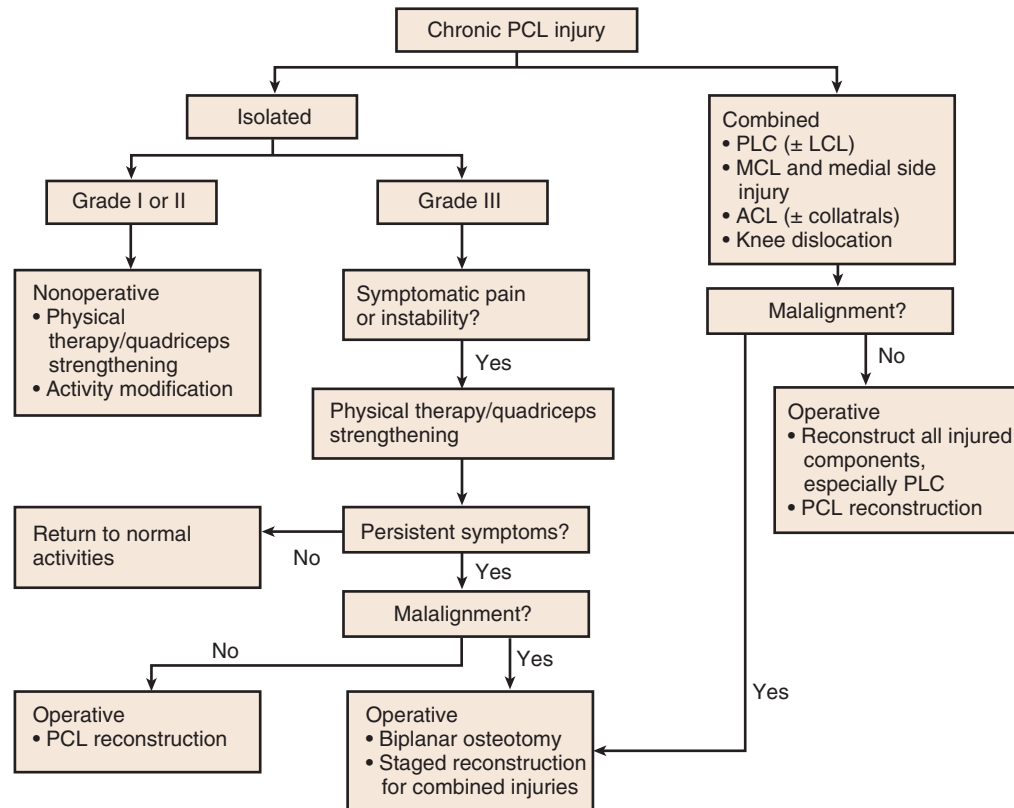
In the first 6 weeks after injury (phase I), rehabilitation is focused on partial weight-bearing, hamstring and gastrocnemius stretching to reduce the posterior pull on the tibia, quadriceps strengthening and prone ROM exercises. In this initial phase, a number of immobilization techniques have been described to decrease stress on the healing ligament. These techniques include bracing the knee locked between 0 and 60 degrees of knee flexion,<sup>63</sup> use of a cylindrical leg cast with a posterior support to prevent posterior displacement of the tibia,<sup>64</sup> and use of a brace with a dynamic anterior drawer to apply an anterior force on the posterior proximal tibia.<sup>65,66</sup> In phase II, 6 to 12 weeks after injury, the focus is on progressive strengthening, reestablishment of full ROM, and improving proprioception. In phase III, 13 to 18 weeks after the injury occurred, the patient is allowed to begin running and to perform sports-specific exercises, with return to sports allowed 4 to 6 months after the initial injury, assuming quadriceps strength is comparable to that of the contralateral leg. It is important to note that this time frame can be significantly accelerated in the case of elite athletes with potential return to sport as early as 6 to 8 weeks but with the expected risk of increased residual joint laxity.<sup>61</sup>

### Operative Treatment

A number of surgical techniques for PCL reconstruction can be considered. Current surgical treatment options include trans-tibial and tibial inlay reconstruction techniques with single- or double-bundle reconstruction and a variety of fixation methods. Several biomechanical and anatomic studies have been published recently investigating the benefits and pitfalls of these techniques.



**Fig. 100.5** Treatment recommendations for acute injuries of the posterior cruciate ligament (PCL). ACL, Anterior cruciate ligament; LCL, lateral collateral ligament; MCL, medial collateral ligament; ORIF, open reduction, internal fixation; PLC, posterolateral corner.



**Fig. 100.6** Treatment recommendations for chronic injuries of the posterior cruciate ligament (PCL). ACL, Anterior cruciate ligament; LCL, lateral collateral ligament; MCL, medial collateral ligament; PLC, posterolateral corner.

However, no consensus currently exists on the best method of PCL reconstruction.

### Transtibial Tunnel Versus Tibial Inlay Techniques

The transtibial technique is a commonly used method of PCL reconstruction. In this technique, the tibial and femoral tunnels are drilled and the graft must make a sharp turn around the “killer turn” as it surfaces from the tibial tunnel and changes direction before entering the knee joint. This acute turn has been implicated as the cause of graft abrasion with subsequent thinning of the graft and eventual graft rupture or excessive laxity.<sup>65</sup> The residual posterior knee laxity observed clinically after traditional transtibial PCL reconstruction techniques may be related to this acute turn. To address the concern of graft attenuation resulting from this tunnel, the tibial inlay technique was developed and reported by Jakob and Rueggsegger,<sup>67</sup> as well as by Berg.<sup>68</sup> In this technique, direct fixation occurs at the tibial attachment site of the PCL, preventing an acute turn as the graft passes from the tibia to the femoral tunnel.

A number of cadaveric biomechanical studies have compared the transtibial and tibial inlay techniques. Although McAllister et al.<sup>69</sup> found no significant differences in mean knee laxities between the tibial tunnel and tibial inlay techniques at time zero, increased laxity was observed with this technique after cyclic loading. Bergfeld et al.<sup>70</sup> assessed anteroposterior laxity in cadaveric knees undergoing tunnel reconstruction or inlay reconstruction. Minimal differences in anteroposterior laxity were observed

in the inlay group when compared with the tunnel group from 30 to 90 degrees of knee flexion and after repetitive loading at 90 degrees of knee flexion. However, evaluation of the grafts after testing demonstrated evidence of graft thinning and attenuation in the tunnel group but not in the inlay group. In a detailed cyclic loading analysis, Markolf and colleagues<sup>71</sup> also evaluated cadaver knees with tibial inlay and transtibial reconstruction. Ten of 31 grafts in the tunnel group failed at the acute angle before 2000 cycles of testing could be completed, whereas all 31 grafts that had been fixed to the tibia with use of the inlay method survived the testing intact. In addition, a significant increase in graft thinning and stretching out was observed in the remaining tunnel grafts that survived testing compared with the inlay grafts.

Thus in vitro analyses comparing the transtibial technique with tibial inlay suggest that although initial knee stability is equivalent, posterior laxity increases with cyclic loading with the transtibial technique when compared with the tibial inlay technique. Attempts have also been made to decrease the effects of the killer turn by reducing the sharp edge at the tibial tunnel exit, but this technique has only been attempted in an animal model.<sup>72</sup> Weimann and colleagues<sup>72</sup> found that rounding the sharp edge of the tibial tunnel decreased graft damage associated with the killer turn in a porcine model of PCL reconstruction. To date, retrospective studies comparing patients undergoing transtibial versus tibial inlay procedures<sup>56,73</sup> have not shown significant differences in subjective outcome or knee laxity measurements.<sup>74,75</sup> Thus although the tibial inlay technique may have



some biomechanical advantages when tested in a cadaveric model, these advantages have yet to be realized in the clinical setting.

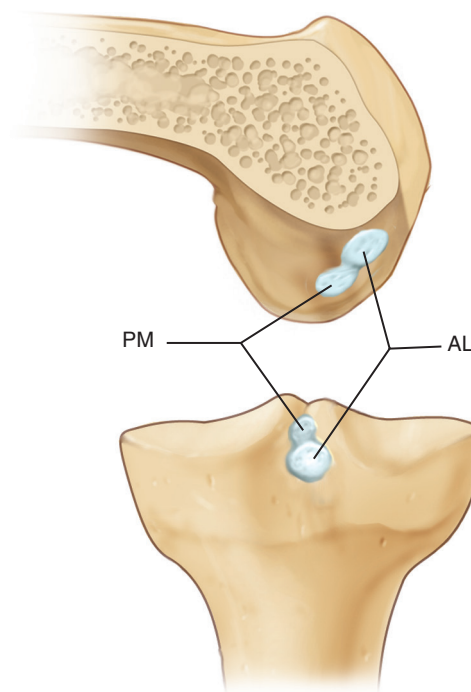
### Single-Bundle Versus Double-Bundle Reconstruction

Controversy also exists regarding the utility of single- versus double-bundle techniques of PCL reconstruction. The native PCL can be divided into an anterolateral (AL) and a PM bundle (Fig. 100.7). The AL bundle is tight in knee flexion and becomes lax in extension, whereas the PM bundle is tight in knee extension and becomes lax in flexion. The AL bundle is larger in cross-sectional area and thus is most commonly reconstructed in single-bundle procedures (Fig. 100.8). Double-bundle PCL reconstructions were proposed to more closely reproduce the anatomy and biomechanical properties of the intact PCL. Biomechanical studies have indicated that the two bundles demonstrate reciprocal tightening during knee ROM and both are active in reducing posterior tibial translation and external tibial rotation, suggesting that both are required for normal knee kinematics.<sup>76,77</sup>

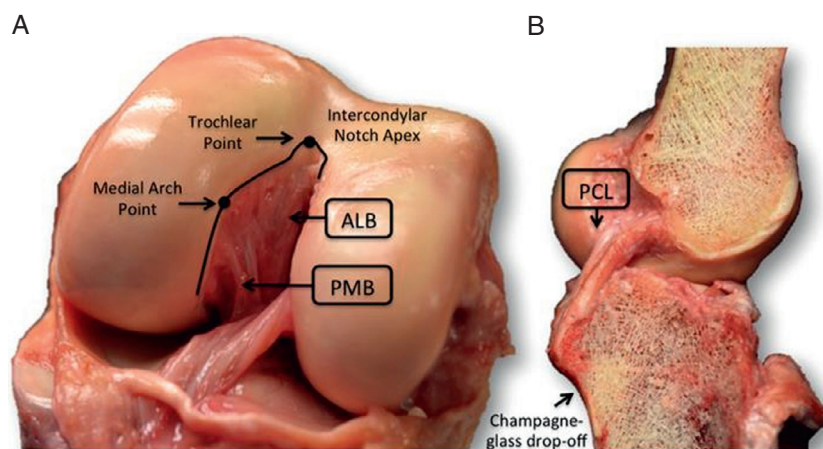
Single- versus double-bundle PCL reconstruction have been compared in several biomechanical studies, and some investigators have suggested improved biomechanics with double-bundle reconstruction.<sup>78,79</sup> Milles et al.<sup>80</sup> compared single-bundle versus double-bundle reconstruction in cadaveric human knees using five different surgical techniques and found increased stiffness and decreased laxity in double-bundle reconstructions at numerous flexion angles. Tsukada et al.<sup>81</sup> compared single AL bundle reconstruction, single PM bundle reconstruction, and double-bundle reconstruction in cadaveric human knees at different angles of knee flexion. The double-bundle reconstruction resisted posterior tibial load better than the AL single bundle at 0 and 30 degrees of knee flexion and better than the PM single bundle at 30, 60, and 90 degrees of knee flexion under the posterior

tibial load, leading the authors to conclude that double-bundle reconstruction reduces laxity in extension.

Additional studies have demonstrated potential drawbacks of double-bundle reconstruction, and some studies have been unable to demonstrate a benefit. Whiddon et al.<sup>82</sup> compared single- and double-bundle tibial inlay reconstruction in a cadaver



**Fig. 100.7** Femoral and tibial insertion sites of the posteromedial (PM) and anterolateral (AL) bundles of the posterior cruciate ligament. (From Chahla J, Nitri M, Civitarese D, et al. Anatomic double bundle posterior cruciate ligament reconstruction. *Arthrosc Tech.* 2016;5(1):149–156.)



**Fig. 100.8** (A) Left knee showing both bundles: anterolateral bundle (ALB) and posteromedial bundle (PMB). The trochlear point is easily identifiable on the distal aspect of the trochlea. The more anterior aspect of the ALB is noted by the trochlear point, whereas its more inferoposterior aspect is delineated by the medial arch point. Likewise, the PMB is located along the wall of the notch and distal to the medial arch point. (B) Profile view of a hemi-sectioned left knee showing the tibial and femoral insertion of the posterior cruciate ligament (PCL). (From Chahla J, Nitri M, Civitarese D, et al. Anatomic double-bundle posterior cruciate ligament reconstruction. *Arthrosc Tech.* 2016;5(1):e149–e156. Elsevier, Inc.)

model and found that the double-bundle technique improved rotational stability and posterior translation in knees with a concomitant PLC injury. However, no advantage was seen with a double-bundle reconstruction when compared with a single-bundle reconstruction with regard to posterior translation with the PLC intact. In addition, excessive rotational constraint was observed at 30 degrees. Wiley and colleagues<sup>83</sup> also observed that although posterior laxity was reduced compared with single-bundle reconstruction, overconstraint at 30 degrees of flexion was seen with double-bundle reconstruction. In a comparison of single-bundle AL reconstruction with double-bundle reconstruction in cadaver knees, Markolf and colleagues<sup>84</sup> found that the addition of a PM bundle reduced laxity from 0 to 30 degrees of flexion but at the expense of increased PCL graft forces. Bergfeld et al.<sup>85</sup> compared single- and double-bundle tibial inlay reconstruction in cadaveric knees using Achilles tendon grafts. No differences in translation between the single- and double-bundle reconstruction were observed at any flexion angle.

A number of clinical studies have shown no significant differences in subjective and objective results between single- and double-bundle PCL graft reconstructions.<sup>86–88</sup> Jain and colleagues performed a retrospective review of 40 patients, 18 of which underwent double-bundle PCL reconstruction and 22 of which underwent single-bundle PCL reconstruction and looked at clinical, functional, and radiologic outcomes at 24 months postoperatively.<sup>8</sup> KT-1000 measurements showed an average side-to-side difference in the double-bundle group of 1.78 mm and 2.4 mm in the single-bundle group ( $P < .04$ ). Stress radiographs showed increased posterior tibial translation in the single-bundle group as compared to the double-bundle group. However, both double-bundle and single-bundle groups showed significant functional improvement in Lysholm and IKDC scores with no statistically significant difference between the groups themselves. Furthermore, Hatayama et al.,<sup>88</sup> in a retrospective clinical study comparing arthroscopic single-bundle and double-bundle PCL reconstruction showed that at 2-year follow-up of 3/10 patients who underwent double-bundle PCL reconstruction showed rupture of the PM bundle at second-look arthroscopy.<sup>89</sup> Based on the literature to date, we believe evidence is lacking to support the routine use of double-bundle reconstruction. However, this type of reconstruction remains a topic of interest and is undergoing continued investigation in the treatment of PCL injuries.

### Graft Choice and Fixation

Graft choice and fixation techniques are also important considerations when discussing treatment options for surgery. Both autograft and allograft tissues have been used for PCL reconstruction. Bone–patellar tendon–bone, hamstring, and quadriceps tendons are common autograft sources. The Achilles tendon, as well as anterior and posterior tibial tendons, are frequently used allografts. Among autografts, bone–patellar tendon–bone grafts have the advantage of bone-to-bone healing in the bone tunnel. In comparison, the tendon portion of the quadriceps graft and both ends of hamstring grafts require tendon-to-bone healing in the bone tunnel, which may have inferior biomechanical properties.<sup>90</sup> Weakening of the quadriceps tendon, which acts

synergistically with the PCL to prevent posterior tibial translation, is a concern with this graft option, as is the variable length of the tendinous portion.<sup>91</sup> Lin et al.<sup>92</sup> performed a retrospective review of 59 patients with isolated PCL injuries who either underwent arthroscopic transtibial single-bundle PCL reconstruction using autologous patellar tendon or hamstring tendon grafts looking for differences in clinical outcomes. At an average of 4 years follow-up, patients who underwent single-bundle PCL reconstruction using bone–patellar tendon–bone grafts demonstrated significantly more kneeling pain, anterior knee pain, squatting pain, posterior drawer laxity, and osteoarthritic change on radiography. Allograft tissue has the advantage of avoiding donor-site morbidity, reducing operating time, and offering improved graft diameter with greater collagen tissue when Achilles and tibialis anterior tendons are used. Pitfalls of allografts include a small risk of disease transmission, cost, and availability. In a survey of orthopaedic surgeons, Dennis et al.<sup>93</sup> reported that allograft Achilles tendon was the most commonly used graft for acute (43%) and chronic (50%) PCL reconstructions. For the reasons previously discussed, we favor the use of Achilles tendon allografts for PCL reconstruction.

A number of biomechanical studies have investigated various graft fixation constructs used in PCL reconstruction. Most recently, Lim et al.<sup>94</sup> compared cross-pin fixation in a porcine model with bone blocks, interference screw fixation with bone blocks, cross-pin fixation of soft tissue with backup fixation, and interference screw fixation of soft tissue with backup fixation on the tibial side using Achilles allograft PCL reconstruction. Although cross-pin fixation with backup fixation had a higher maximum failure load and stiffness, tendon graft displacement was increased compared with bone-block fixation. Gupta and colleagues compared bioabsorbable to metallic screws for inlay fixation and found no difference in failure load or linear stiffness.<sup>95</sup> Markolf and colleagues<sup>96</sup> demonstrated the importance of bone-block position and orientation within the tibial tunnel; they found that positioning the bone–patellar tendon–bone graft flush with the posterior tunnel opening with the graft oriented so the bone block faced anteriorly in the tibial tunnel was the position with the best biomechanical properties. Margheritini et al.<sup>97</sup> found that combining distal and proximal tibial fixation resulted in significantly less posterior tibial translation and more closely restored intact PCL in situ forces at 90 degrees than did reconstruction with distal fixation.

### High Tibial Osteotomy for Chronic PCL Injuries

A chronic isolated PCL injury results in posterior translation of the tibia and external rotation of the tibia in relation to the femur. These anatomic changes result in increased forces and subsequent development of OA in the medial and patellofemoral compartments.<sup>98–100</sup> Additionally, a chronic combined PCL and PLC injury can result in chronic posterolateral instability and varus malalignment associated with bony deformity, lateral soft tissue deficiency, and hyperextension and external rotation as a result of PLC deficiency (i.e., triple varus).<sup>101</sup> In cases of chronic PCL or combined PCL/PLC injury with resultant varus malalignment and posterior or posterolateral instability, soft-tissue procedures alone may be insufficient, whereas performance of a

high tibial osteotomy (HTO) prior to soft-tissue reconstruction may improve outcomes by decreasing forces across the lateral supporting structures of the knee.

A medial opening wedge HTO can improve alignment and decrease instability by addressing both coronal and sagittal malalignment. In addition to correcting varus malalignment, a medial opening wedge osteotomy with the anteromedial gap equal or larger than the posterior medial gap increases the posterior tibial slope<sup>102</sup> and thus decreases the posterior resting position of the tibia.<sup>103</sup> In contrast, a lateral closing wedge osteotomy may decrease posterior tibial slope, consequently increasing the posterior resting position of the tibia,<sup>104</sup> and thus may not be appropriate in the knee with a PCL injury. Several investigators have demonstrated a concomitant increase in posterior tibial slope with opening wedge HTO.<sup>105,106</sup> Specifically, Noyes et al.<sup>102</sup> calculated that for each increase of 1 mm in the anterior gap, an increase of 2 degrees occurs in the posterior tibial slope. In the coronal plane, in the absence of medial compartment OA, the osteotomy should result in the mechanical axis crossing the center of the knee. If medial compartment joint space narrowing is present, some authors have recommended valgus hypercorrection with the mechanical axis crossing lateral to the center of the knee.<sup>101</sup>

Although satisfactory long-term outcomes have been observed in patients undergoing HTO for medial compartment OA,<sup>106</sup> studies reporting results of HTO specifically for chronic PCL deficiency or combined PCL/PLC insufficiency are limited and heterogeneous in patient population, duration of follow-up, and outcomes. In a case series of 17 HTOs for symptomatic hyperextension varus thrust that included four patients with isolated PCL injuries and seven with combined PCL and posterolateral ligament injuries, improved subjective activity scores were observed postoperatively at a mean follow-up of 56 months.<sup>105</sup> Similarly, Badhe and Forster<sup>107</sup> reported the results of HTO with or without ligament reconstruction in 14 patients with knee instability and varus alignment, including nine patients with PLC or combined PCL injury. The mean time from injury to HTO was 8.3 years. Although the mean Cincinnati Knee Score improved from a mean preoperative score of 53 to a mean postoperative score of 74, no patients were able to participate in competitive sports, and more than 30% had continued knee pain at follow-up. Thus although biomechanical studies suggest that HTO may improve alignment and stability in patients with chronic PCL insufficiency or combined PCL/PLC injury with varus malalignment and instability, data on outcomes in this patient population are currently minimal.

## Authors' Preferred Technique

### **Posterior Cruciate Ligament Reconstruction**

As discussed previously, many techniques have been described for PCL reconstruction. We prefer the tibial inlay method of reconstruction with an Achilles tendon allograft. The tibial inlay approach avoids the "killer turn" that may predispose the graft to stretch out and failure. We also believe that the acute angle of the graft as it enters the notch in the transtibial technique can make tensioning more difficult. Some surgeons are concerned that the posterior approach to the tibia needed for the open inlay procedure, which may involve a change to the prone position, is more technically demanding. We believe that with experience these challenges are easily overcome and that this technique leads to better biomechanical stability.

Previous studies have demonstrated that the AL bundle is the most important component of the native PCL. The AL bundle has a higher cross-sectional area and is stronger than the PM bundle. Therefore the goal of a single-bundle reconstruction is to recreate the native AL bundle. However, it should be remembered that the footprint of the native PCL is much larger than the typical drill used to create the femoral tunnel, and thus the surgeon must choose which portion of the PCL to reconstruct. Clinical studies comparing single- versus double-bundle reconstructions have not found any significant differences in patient outcome scores. Multiple options are available for graft tissue, and no study has conclusively demonstrated a superior graft. We use an Achilles tendon allograft for most of our PCL reconstructions. We prefer to use the Achilles tendon because of its size, strength, and versatility. For the aforementioned reasons, we prefer to use a single-bundle tibial inlay technique with use of an Achilles allograft for PCL reconstruction. This procedure is described in the following sections.

#### **Graft Preparation**

The soft-tissue portion of the Achilles allograft is sized for a 10-mm bone tunnel. The bone plug is then fashioned into a trapezoidal shape 25 mm in length and 13 mm in width. The bone plug is predrilled and tapped for a 6.5-mm cancellous screw. The bone plug is drilled with a 4.5-mm drill from the cancellous to cortical

surface to protect the soft tissue and is then tapped with a 6.5-mm tap. A 6.5-mm cancellous screw, approximately 35 mm in length, and a metal washer are then placed into the bone plug from the soft tissue/cortical surface to the cancellous surface. The screw is placed so that the tip is 5 mm past the cancellous surface to facilitate later tibial fixation. A running locking stitch is then placed along approximately 30 mm of tendon using a no. 2 braided polyester suture, which tabularizes the graft to aid in passage through the femoral tunnel.

#### **Arthroscopy/Femoral Tunnel**

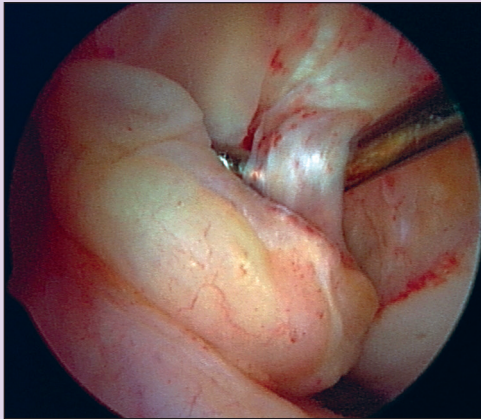
For the arthroscopic portion of the case, the patient is laid supine on the operative table. We recommend that the patient be intubated to protect the airway during position changes. A complete examination is performed after inducement of anesthesia prior to placement of a tourniquet. It is very important to evaluate for both PCL and associated capsuloligamentous injuries at this time. A thigh tourniquet is then placed but not inflated. A routine diagnostic arthroscopy is performed, and any meniscal or chondral injuries are treated at this time. The ACL may appear lax because of posterior tibial subluxation and should tighten with an applied anterior drawer. The PCL is then examined, and often, the ligament is lax or stretched out rather than frankly torn (Fig. 100.9). Once incompetence of the PCL has been confirmed, the residual PCL tissue is removed with a shaver and hand-operated punches. If the ligaments of Humphrey and Wrisberg are present, they are preserved if possible. The native footprint is preserved as a guide for femoral tunnel placement. Our goal of reconstruction is to restore the AL bundle of the PCL. The tunnel is placed in the distal and anterior portion of the native PCL footprint. A small medial incision is made through the skin and then through the medial retinaculum, which facilitates optimal drill-guide placement with the tunnel oriented slightly posteriorly. The medial articular margin is used as a landmark for the guide. An outside-in arthroscopic guide is used to establish the tunnel position, and a femoral guide pin is placed. The femoral tunnel is created with a cannulated drill over the guide pin (Fig. 100.10).



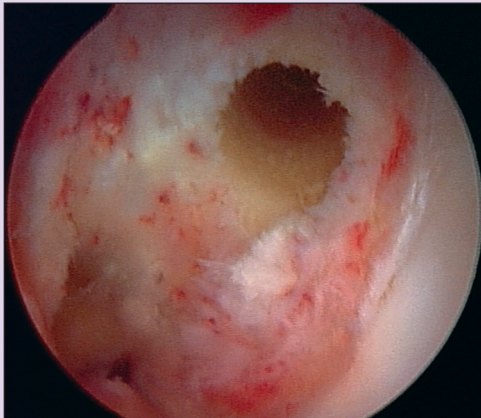


## Authors' Preferred Technique

### Posterior Cruciate Ligament Reconstruction—cont'd



**Fig. 100.9** An arthroscopic view of a posterior cruciate ligament tear.



**Fig. 100.10** An arthroscopic view of the femoral tunnel created in the footprint of the native anterolateral bundle with a cannulated drill over a guide pin.

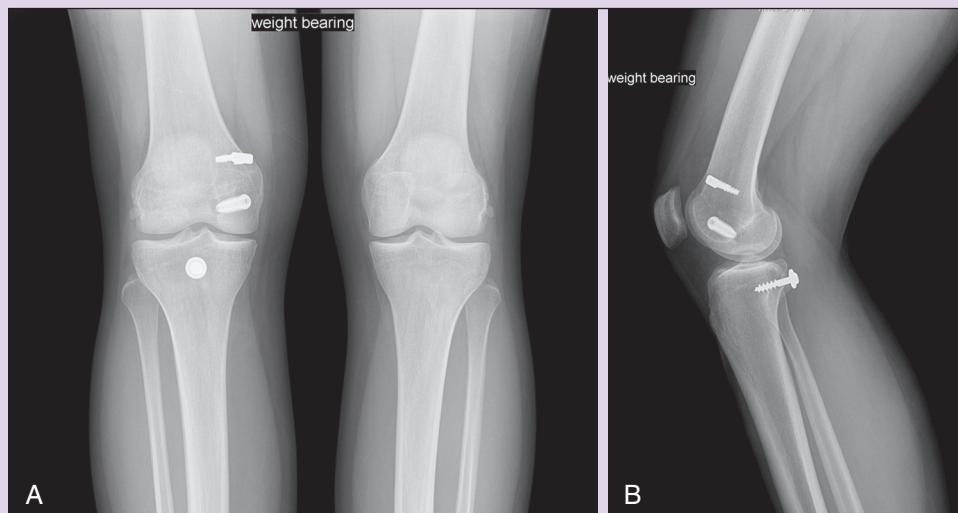
The drill size is determined by the size of the graft and is typically 10 mm in width. An 18-gauge wire loop is passed through the femoral tunnel from the outside and positioned in the posterior notch to be retrieved later for graft passage.

#### Tibial Inlay

The patient is then rotated into the prone position in a sterile fashion in preparation for the tibial inlay portion of the case. The extremity is exsanguinated and a tourniquet is inflated. The PM exposure to the tibia, as described by Burks,<sup>108</sup> is then performed. The skin incision is a gentle curve with a horizontal end at the medial popliteal crease and vertical limb overlying the medial aspect of the gastrocnemius. Dissection is carried down the investing fascial layer, which is incised over the medial head of the gastrocnemius. The medial sural cutaneous nerve can be at risk but typically perforates the fascia distal to the horizontal limb of the incision. The medial border of the medial gastrocnemius is identified. The interval between the medial gastrocnemius and semimembranosus tendon is developed. Blunt dissection is performed down to the joint capsule. The medial head of the gastrocnemius is then retracted laterally with a blunt-tipped retractor, which protects its motor branch and neurovascular structures. At this point the posterior proximal tibia and posterior femoral condyles are palpated and a vertical incision is made through the posterior capsule. The posterior notch and tibial attachment of the PCL should now be exposed and the tibial insertion site of the native PCL is prepared for placement of the graft. Typically two prominent processes are found on the medial and lateral borders of the PCL that can be palpated. The insertion site is resected using osteotomes, a rongeur, and/or a burr. A graft recipient site is created that will anatomically accommodate the previously prepared graft. The bone graft is then placed into the site and secured with a 6.5-mm cancellous screw and washer (Fig. 100.11). The sutures in the tendinous portion of the graft are shuttled through the femoral tunnel using the previously placed wire loop. After the sutures are passed, the capsule is repaired. The tourniquet is deflated and hemostasis is achieved. The wound is irrigated and closed in layers.

#### Graft Tensioning

After wound closure the patient is again returned to the supine position in a sterile fashion. The arthroscope is placed back into the knee and the graft is inspected, entering the femoral tunnel (Fig. 100.12). The knee is cycled several



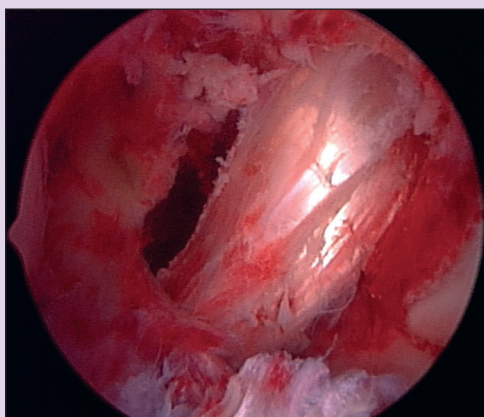
**Fig. 100.11** (A) Anteroposterior and (B) lateral radiographs of the knee after completion of single-bundle posterior cruciate ligament reconstruction.



## Authors' Preferred Technique

### Posterior Cruciate Ligament Reconstruction—cont'd

times, which allows the surgeon to evaluate ROM and helps apply tension to the graft. Tension is applied to the graft in 70 to 90 degrees of flexion with an anterior drawer force placed on the proximal tibia. A  $9 \times 25$ -mm soft-tissue interference screw is then used to fix the graft in the femoral tunnel. A staple is then placed over the soft tissue portion of the graft into the medial femoral condyle to augment fixation. The arthroscopic portals and the incision for the femoral tunnel are closed in the standard fashion.

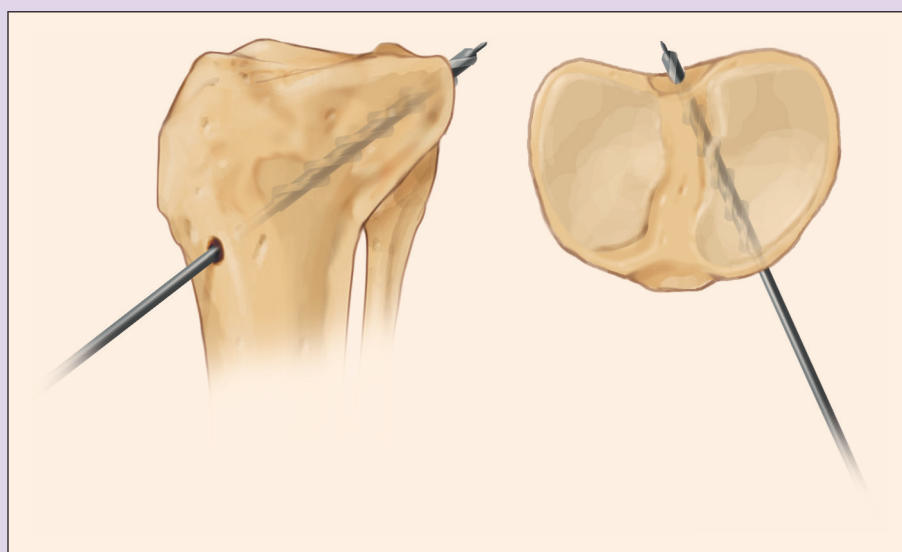


**Fig. 100.12** An arthroscopic view of a completed posterior cruciate ligament reconstruction.

### Transtibial Technique and Double-Bundle Reconstruction

Alternatively, PCL reconstruction can be achieved via an arthroscopic transtibial technique. The patient is positioned supine. A detailed examination is performed after inducement of anesthesia followed by arthroscopic assessment of the knee joint to confirm the extent of the injury and to assist with the repair or reconstructive procedure. The tibial footprint is prepared via an accessory anteromedial portal, and occasionally with use of a 70-degree arthroscope. A tibial tunnel is then created from the anteromedial tibia and directed posteriorly to the native PCL tibial attachment (**Fig. 100.13**). If a single-bundle procedure is performed, care is taken to place the single guidewire in the center of the tibial footprint via direct visualization and/or radiographic guidance with use of a guide. If a double-bundle reconstruction is performed, two guidewires are placed and confirmed radiographically, with the AL guidewire being more lateral and distal and the PM guidewire being more medial and proximal. The tunnel(s) is (are) reamed under power to the posterior cortex and then completed by hand with direct visualization.

After the tibial tunnels are completed, attention is focused on creating the femoral tunnels. The femoral insertion site anatomy is identified, and the appropriate tunnel position is marked for a single-bundle reconstruction or double-bundle reconstruction. The lateral portal is enlarged, and the knee is hyperflexed to drill the femoral tunnel(s) (**Fig. 100.14A**). One or two grafts are used depending on whether a single-bundle or double-bundle reconstruction is being performed. These grafts are passed anterograde through the tibial tunnel (**Fig. 100.15**) and subsequently retrograde into the femur. The grafts are fixed on the femoral or tibial side and then tension is applied to the other side of the graft and it is fixed. Tension is applied to the AL bundle and it is fixed at 90 degrees of flexion, whereas tension is applied to the PM bundle and it is fixed at 30 degrees of flexion.

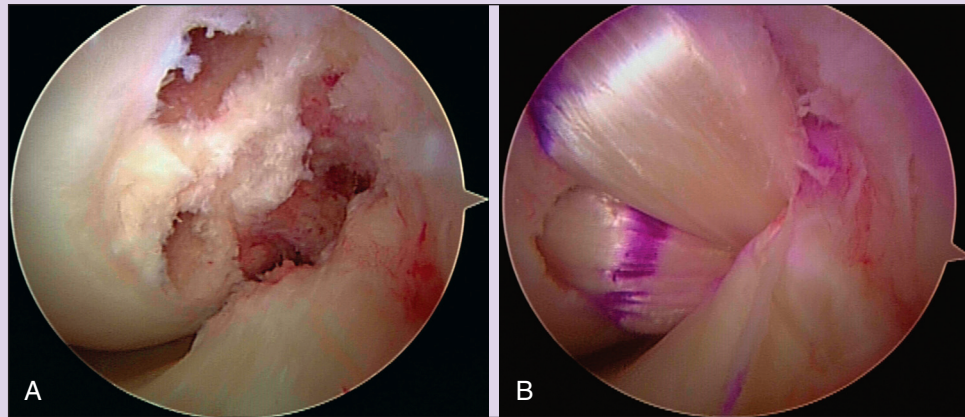


**Fig. 100.13** Posterior cruciate ligament tibial guidewire placement and drilling. (Modified from Miller MD, Harner CD, Koshiwaguchi S. Acute posterior cruciate ligament injuries. In: Fu FH, Harner CD, Vince KG, eds. *Knee Surgery*. Vol. 1. Baltimore: Williams & Wilkins; 1994.)

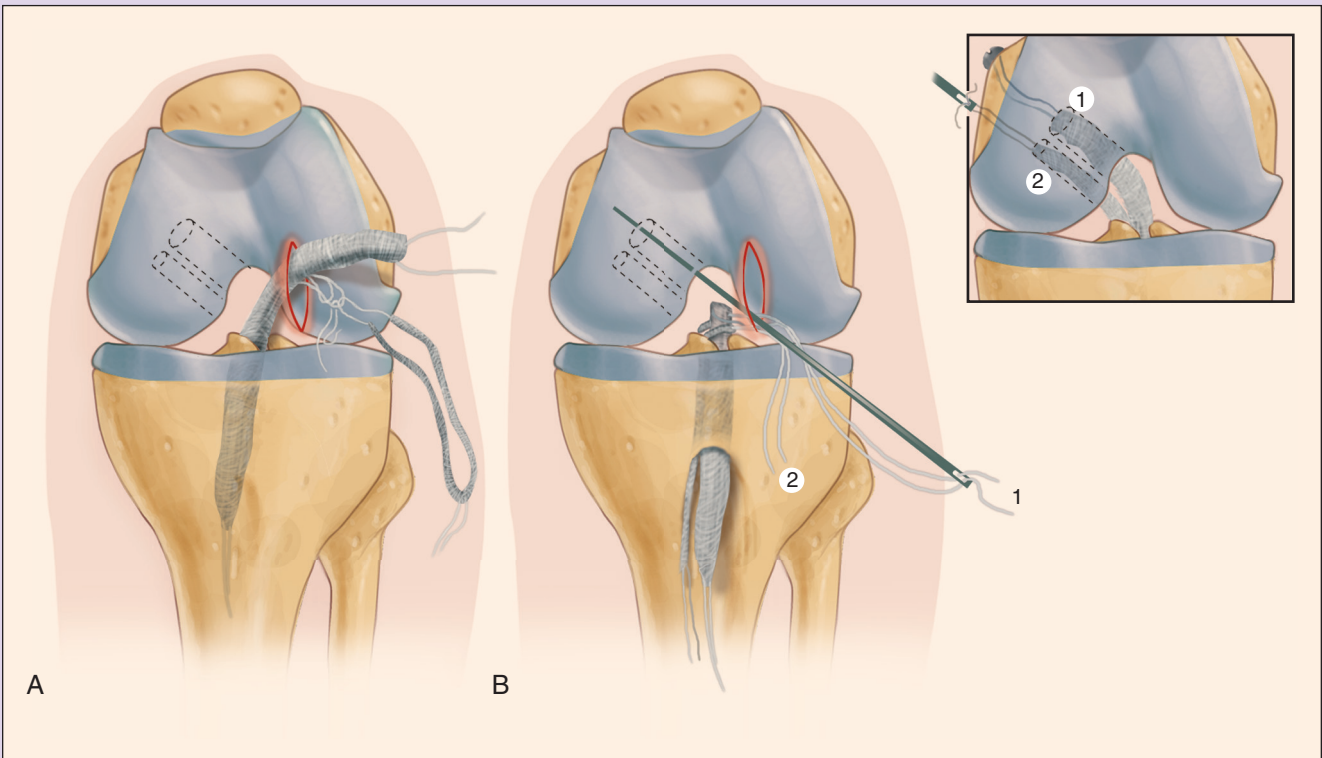


### Authors' Preferred Technique

#### Posterior Cruciate Ligament Reconstruction—cont'd



**Fig. 100.14** Positioning of femoral tunnels for double-bundle reconstruction. (A) Femoral tunnel position for anterolateral and posteromedial bundles. Note that the anterolateral bundle is more anterior. (B) Double-bundle reconstruction with tibialis anterior allografts.



**Fig. 100.15** Graft placement. (A) The graft for the anterolateral bundle (inset, 1) and a second graft for the posteromedial bundle (inset, 2) are passed in anterograde fashion through the tibial tunnel. (B) The grafts are then fixed to corresponding femoral tunnels. (Modified from Petrie RS, Harner CD. Double bundle posterior cruciate ligament reconstruction technique: University of Pittsburgh approach. *Op Tech Sports Med.* 1999;7:118–126.)

## POSTOPERATIVE MANAGEMENT

Although rehabilitation after ACL reconstruction has been investigated by numerous authors, postoperative management after PCL reconstruction has been studied less extensively. Because PCL reconstruction has not been as successful as ACL reconstruction in restoring objective stability, many surgeons recommend a more conservative postoperative course. No level I studies have been performed to compare different protocols, and in a recent review of the literature, Pierce et al.<sup>61</sup> found that currently no consensus exists regarding a set of optimal guidelines. In general, rehabilitation should focus on quadriceps strengthening and regaining ROM while avoiding posterior tibial translation, which places stress on the graft. These goals are usually achieved by initial immobilization and avoidance of active hamstring forces, limited ROM, and progressive weight bearing and strengthening based on the time from surgery and the patient's progress.

The aim of reconstruction and rehabilitation is to help the patient return to previous levels of function. The goal of most protocols is to achieve this goal around 9 months after surgery. Postoperative protocols use functional progression to determine the patient's advancement. Functional status is determined using a combination of subjective patient assessment and objective data. These data include joint stability, ROM, effusion, proprioception deficits, muscle strength, and gait abnormalities. Careful monitoring of these factors during rehabilitation can help the surgeon identify and manage potential complications early during the postoperative course. It is important for the patient to understand that strict compliance with the postoperative protocol is critical to a good outcome.

Traditionally, PCL protocols are divided into specific phases, and advancement is based on time from surgery and patient progression. The nature of these phases varies among surgeons, but many similarities exist. The first phase usually emphasizes protecting the graft from stress, ROM within limits, non-weight bearing, effusion prevention, and reactivation of the quadriceps. The next phase focuses on regaining full ROM, advancement of weight bearing, and low-impact strengthening. The length and method of immobilization have varied among different studies. The initiation of ROM exercises and degrees of motion allowed during the early phases also differs among studies. Progression through the subsequent phases depends on patient function. Factors that influence progression include strength, stability, endurance, and agility. Patients are allowed to return to play when they regain appropriate quadriceps strength and painless active ROM, which, as previously stated, is usually at around 9 months after surgery.

To regain strength and proprioception, most protocols include open and closed kinetic chain exercises. For this reason it is important that open- and closed-chain exercises be part of the PCL rehabilitation program. Open-chain knee flexion resulting from hamstring contraction causes significant posterior translational forces across the tibia and should be avoided during early phases of rehabilitation.<sup>109</sup> Open-chain knee extension can protect the graft by producing anteriorly directed shear forces across the tibia, and most investigators advocate that these forces

be initiated early.<sup>61,109</sup> Closed kinetic chain exercises at low arcs of motion have also been shown to decrease posterior shear forces and are included in PCL rehabilitation. We use the following protocol.

## RESULTS OF OPERATIVE TREATMENT

### Isolated Posterior Cruciate Ligament Injuries

The results of isolated PCL reconstruction have been evaluated in many published studies. Although most patients demonstrate



### Authors' Preferred Technique

#### Postoperative Management

##### A Acute Immediate Postoperative Phase (Early Protection Phase)

- Bracing: After surgery, the hinged brace is locked at zero
- ROM: passive ROM (initiated after 2 to 3 weeks of immobilization in extension)—patient-assisted tibial lift into flexion (0 to 70 degrees) with passive knee flexion
- Exercises:
  - Quadriceps isometrics, straight leg raise—adduction, abduction proximal weight

##### B Acute Phase (Maximal Protection Phase)

- Goals:
  - Minimize external forces to protect the graft
  - Prevention of quadriceps atrophy
  - Control postsurgical effusion
- Weight bearing: weight bearing as tolerated with an assistive device
- ROM: as tolerated to 90 degrees
- Exercises:
  - Continue isometric exercises and quadriceps strengthening
  - Closed kinetic chain mini-squat, shuttle, bike
  - Open kinetic chain knee extension (60 degrees to 0 degrees)
  - Proprioception training
  - Weight shifts
- Brace: Fit with a functional brace at 4 to 6 weeks after surgery

##### C Progressive Range of Motion/Strengthening Phase

- Weight bearing: weight bearing as tolerated without an assistive device
- ROM: as tolerated to 125 degrees flexion
- Exercises:
  - Continue quadriceps strength training
  - Begin isotonic quadriceps strength exercises
  - Leg press (0 to 60 degrees)
  - Step-ups
  - Sport-cord progression program
  - Rowing, NordicTrack
  - Initiate closed kinetic chain terminal knee extension

##### D Functional Activity Phase

Few scientific data are available to help determine the best method of rehabilitation as the patient transitions into functional stages, and thus progression at this level is determined by the patient's tolerance to exercise and level of function. The evaluation of power and endurance that have been used for ACL programs should theoretically measure total length, strength, and endurance and can be used for the PCL reconstructed knee. As previously stated, the anticipated return to previous activity after PCL reconstruction is anticipated to take approximately 9 months.



improvement, some patients report persistent knee symptoms and have residual laxity that may inhibit return to preinjury-level activities.<sup>56,65,88,89,110–122</sup> The main limitation of these studies is the low incidence of isolated PCL injury requiring reconstruction, which results in a small sample size. Many of these studies are also retrospective case series that differ in operative technique, outcome measurements, and postoperative treatment, making direct comparison between studies difficult. A summary of these studies can be found in Table 100.1.<sup>112,120,121,123–130</sup>

Hermans et al.<sup>112</sup> evaluated 25 patients who underwent isolated AL bundle PCL reconstructions at an average follow-up of 9.1 years. They used an arthroscopic transtibial, single-bundle approach with use of various graft types. The final International Knee Documentation Committee (IKDC), Lysholm, and functional Visual Analog Scale scores were significantly better than preoperative scores, but only 41% of patients had normal or near-normal clinical findings according to the IKDC guidelines. This finding was mostly attributed to residual laxity with a mean side-to-side difference of 4.7 mm by Telos stress radiographs at final follow-up. Preoperative symptomatic instability greater than 1 year and chondrosis at the time of surgery correlated with poorer subjective outcomes. These findings confirm that although isolated PCL reconstruction often results in improved functional outcome, many patients have residual laxity. Lahner et al.<sup>121</sup> prospectively followed up on 33 patients with chronic symptomatic PCL injuries who underwent isolated single-bundle transtibial reconstruction. These patients were followed up over 2 years, and during this time, their IKDC scores improved from 41.8 to 69.5, and 72.8% regained normal to near-normal knee function. In this study, nine patients (27.4%) showed no improvement in knee function postoperatively. Chen and Gao<sup>120</sup> evaluated a transtibial double-bundle reconstruction using a suture suspension technique at a minimum follow-up of 2 years. In their series of 19 patients, 78.9% regained normal knee function and 15.8% regained near-normal knee function. These rates are significantly higher than those reported in other studies and may be attributable to higher preoperative scores. The patients in this series had an average preoperative IKDC score of 65.6 and less posterior laxity compared with subjects in other studies.

### Transtibial Versus Tibial Inlay

Several case series have compared the outcomes of transtibial and tibial inlay methods of PCL reconstruction. No significant difference in clinical outcome scores between the two techniques have been identified. The details of these studies are outlined in Table 100.2. MacGillivray et al.<sup>56</sup> evaluated 20 patients who underwent reconstruction of an isolated PCL injury. The mean follow-up was 5.7 years. Thirteen patients underwent transtibial reconstruction and seven underwent tibial inlay, all with a single-bundle graft. These investigators found that the posterior drawer improved in 57% of patients in the inlay group and in 38% in the transtibial group. No significant difference in Tegner, Lysholm, or American Academy of Orthopaedic Surgeons knee scores was found between the two groups. The investigators concluded that neither method predictably restores original laxity and that there was no difference in outcome scores.

Seon and Song<sup>73</sup> retrospectively reviewed 21 isolated transtibial and 22 tibial inlay reconstructions. They found a significant improvement in Lysholm knee scores in both groups but no intergroup differences. Postoperative Tegner scores were also improved in both groups. Final follow-up found normal or grade I laxity on posterior drawer in 19 patients in the transtibial group and in 20 patients in the inlay group. Mean side-to-side differences were also improved, with no significant difference between the two groups. The authors concluded that the transtibial and inlay techniques resulted in relatively good clinical and objective outcomes and are both satisfactory options for reconstruction. Kim et al.<sup>115</sup> compared three different PCL reconstruction techniques. Twenty-nine patients underwent single- or double-bundle arthroscopic tibial inlay reconstruction or transtibial single-bundle reconstruction. These investigators found a significant difference in postoperative posterior tibial translation when the double-bundle tibial inlay group was compared with the single-bundle transtibial group, with less translation in the former. Although they noted a difference in translation, they found no significant difference in postoperative Lysholm scores or ROM. Song and colleagues retrospectively reviewed 36 isolated transtibial and 30 isolated tibial inlay reconstructions at an average follow-up of 148 months. Both groups showed no significant differences in terms of Lysholm knee scores, Tegner scores, or residual posterior laxity. Knees from both groups demonstrated development of OA on radiograph with no significant difference in prevalence.<sup>75</sup>

### Single Versus Double Bundle

A number of in vivo studies have compared single-bundle and double-bundle PCL reconstructions. Yoon et al.<sup>131</sup> prospectively followed up 53 patients who underwent single- or double-bundle reconstruction. All reconstructions were performed with a transtibial approach using Achilles tendon allograft and had a minimum of 2-year follow-up. At final follow-up, patients were evaluated for ROM and posterior stability using stress radiography and by subjective knee scoring. The authors found no significant difference in ROM, Tegner activity scores, Lysholm scores, and IKDC evaluation. The only difference that could be identified was in posterior laxity. Both groups had improved stability postoperatively; however, the double-bundle group had less posterior translation with a difference of 1.4 mm, which was statistically significant. Although they did have less instability on objective testing, the patients' clinical outcomes as measured by subjective scoring were the same in both groups. These findings were later corroborated by a study by Li et al. who prospectively followed 46 patients who underwent single- or double-bundle PCL reconstructions using tibialis allograft via a transtibial approach. In this study, the authors found no significant difference in Lysholm scores, or Tegner activity scores but the double-bundle group showed a statistically significant difference in residual posterior laxity compared to the single-bundle group (2.2 mm vs. 4.1 mm).<sup>132</sup>

Several studies to date have shown no significant differences in subjective or objective results between single- and double-bundle PCL graft reconstructions. Wang et al.<sup>127</sup> prospectively compared single- and double-bundle PCL reconstruction using



TABLE 100.1 Results of Isolated Posterior Cruciate Ligament Reconstruction

| Study                                                  | No. of Patients                                             | Age (Years) | Follow-up (Years) | Chronicity/Grade                                                                                     | Graft Type                                                                           | Surgical Technique         |
|--------------------------------------------------------|-------------------------------------------------------------|-------------|-------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------|
| Lahner et al. (2012), <sup>114</sup><br>prospective    | 33                                                          | 32.5        | 2                 | Chronic<br>13 grade II<br>20 grade III                                                               | Hamstring autograft                                                                  | Transtibial, single bundle |
| Hermans et al. (2009), <sup>105</sup><br>retrospective | 25                                                          | 30.8        | 9.1               | 7 acute, 18 chronic, grade II or more                                                                | 9 BPTB autograft<br>7 quadrupled hamstring autograft<br>8 double hamstring autograft | Transtibial, single bundle |
| Chen and Gao (2009) <sup>113</sup>                     | 19                                                          | 39          | Minimum of 2      | Unknown<br>Average time 14 months<br>15 grade II<br>4 grade III                                      | 8-strand hamstring autograft                                                         | Transtibial, double bundle |
| Chan et al. (2006), <sup>116</sup><br>prospective      | 20                                                          | 29          | 3.3               | Unknown<br>Average time 4 months<br>All grade III                                                    | Quadrupled hamstring autograft                                                       | Transtibial                |
| Sekiya et al. (2005), <sup>117</sup><br>retrospective  | 21                                                          | 38          | 5.9               | 5 acute<br>16 chronic<br>All grade III                                                               | Achilles allograft                                                                   | Transtibial                |
| Ahn et al. (2005), <sup>118</sup><br>retrospective     | Group I: 18<br>Group II: 18                                 | 30<br>31    | 2.9<br>2.3        | All chronic<br>11 grade II<br>7 grade III<br>10 grade II<br>8 grade III                              | Hamstring autograft<br>Achilles allograft                                            | Transtibial<br>Transtibial |
| Jung et al. (2004), <sup>119</sup><br>retrospective    | 12                                                          | 29          | 4.3               | Unknown<br>Average time 5.4 months<br>Range 1–10 months                                              | Patellar tendon autograft                                                            | Tibial inlay               |
| Wang et al. (2003), <sup>120</sup><br>retrospective    | 30                                                          | 32          | 3.3               | 13 acute<br>17 chronic<br>All grade III                                                              | Mixed                                                                                | Transtibial                |
| Deehan et al. (2003), <sup>121</sup><br>prospective    | 27                                                          | 27          | 3.3               | All chronic (16 patients 4–12 months after injury, 11 patients >1 year)<br>Grade II and III injuries | Hamstring autograft                                                                  | Transtibial                |
| Chen et al. (2002), <sup>122</sup><br>retrospective    | Group A (quad tendon): 22<br>Group B (hamstring tendon): 27 | 29<br>27    | 2.5<br>2.2        | All grade III<br>12 acute<br>10 chronic<br>16 acute<br>11 chronic                                    | Quadriceps tendon autograft<br>Quadrupled hamstring autograft                        | Transtibial                |
| Mariani et al. (1997), <sup>123</sup><br>retrospective | 24                                                          | 26          | 2.2               | All chronic                                                                                          | Patellar tendon autograft                                                            | Transtibial                |

A, Abnormal; *bio*, biologic; BPTB, bone–patellar tendon–bone; IKDC, International Knee Documentation Committee; IS, interference screw; KT, KT-1000 testing; N, normal; NA, not available; NN, near-normal; OAK, Orthopädische Arbeitsgruppe Knie; SA, severely abnormal.

| Fixation                                                                                                                     | Subjective Outcome                                                                                                                                | Instrumented Laxity                                                                                                                                                                          | Posterior Drawer                                                                                     | Miscellaneous                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tibia: bio IS + button<br>Femur: FlippTack<br>Tibia: IS<br>Femur: IS                                                         | IKDC: 69, 72.8% N/NN<br>Tegner: 5.9<br><br>IKDC: 65<br>Lysholm 75<br>Tegner 5.7                                                                   | Telos radiograph: 5 mm<br><br>KT posterior drawer: 4.7 mm<br>Side-side difference: 2.1 mm                                                                                                    | Grade I: 25<br>Grade II: 8<br><br>Grade 0: 2<br>Grade I: 15<br>Grade II: 5                           | Minor extension deficit 3–5 degrees in only 2 patients<br>11 patients with residual positive quadriceps active test<br>IKDC N to NN in only 41%<br>Poorer results in chronic injuries<br>Average preoperative IKDC score 65.6 |
| Suture suspension                                                                                                            | Lysholm: 92.1<br>IKDC: 92.1, 78.9% N, 15.8% NN                                                                                                    | KT posterior drawer: 9.4 preoperative to 1.0 postoperative<br>Stress radiograph: 2.0                                                                                                         | Grade 0: 17<br>Grade I: 1<br>Grade II: 1                                                             |                                                                                                                                                                                                                               |
| Tibia: bio IS + screw and washer<br>Femur: bio IS + washer<br>Tibia: screw and washer<br>Femur: metal IS                     | Lysholm: 93<br>Tegner: 6.3<br>IKDC: 85% N/NN<br><br>IKDC knee function: 57% N/NN, 43% A/SA<br>IKDC activity level: 62% N/NN, 38% A/SA             | Average postoperative KT posterior drawer: 3.8 mm<br><br>KT posterior drawer: 4.5 mm<br>KT side-side difference: 1.96 mm                                                                     | Grade I: 16<br>Grade II: 3<br>Grade III: 1<br><br>IKDC acute/subacute: 75% N/NN<br>Chronic: 40% N/NN | 18/20 showed no radiographic deterioration<br>3 patients had stiffness<br>Acute/subacute group had significantly better IKDC and KT-1000 than chronic group                                                                   |
| Femur: IS and screw and washer<br>Tibia: IS and screw and washer                                                             | Lysholm: 90<br>IKDC: 16 N/NN, 2 A<br><br>Lysholm: 85<br>IKDC: 14 N/NN, 3 A, 1 SA                                                                  | Telos stress radiograph posterior displacement: 2.2 mm<br>2.0 mm                                                                                                                             | NA<br><br>NA                                                                                         | No difference in outcome between groups                                                                                                                                                                                       |
| Femur: IS<br>Tibia: screw and washer                                                                                         | OAK score: 92.5; 7 excellent, 4 good<br>IKDC: 11/11 were N/NN                                                                                     | Stress radiographs: 3.4 mm side-side difference; KT side-side difference, 1.8 mm                                                                                                             | NA                                                                                                   |                                                                                                                                                                                                                               |
| Femur: IS; tibia: screw and post                                                                                             | Lysholm: 92 (24 excellent/good, 6 fair/poor)<br>Tegner: 4.5                                                                                       | NA                                                                                                                                                                                           | Grade I: 16<br>Grade II: 12<br>Grade III: 3                                                          | Significant correlation between poor results and more chronic injuries                                                                                                                                                        |
| Femur: IS<br>Tibia: IS                                                                                                       | Lysholm: 94<br>IKDC: 25 N/NN, 2 A/SA                                                                                                              | KT side-side difference <2 mm: 17 patients<br>3–4 mm, 6 patients                                                                                                                             | Grade 0/I: 23<br>Grade II: 1                                                                         | No correlation between time from injury to surgery and outcome                                                                                                                                                                |
| Tibia: suture and post<br>Femur: metal IS<br>Tibia: screw and washer<br>Femur: screw and washer<br>Tibia and femur: metal IS | Lysholm: 90.63<br>IKDC: 18 N/NN, 4 A/SA<br><br>Lysholm: 91.44<br>IKDC: 22 N/NN, 5 A/SA<br><br>Lysholm: 94<br>Tegner: 5.4<br>IKDC: 19 N/NN, 5 A/SA | Average postoperative KT posterior drawer: 3.72 mm<br>Average postoperative KT posterior drawer: 4.11 mm<br><br>KT side-side differences: 6, 0–2 mm<br>13, 3.5 mm<br>3, 6–10 mm<br>2, >10 mm | NA<br><br>NA<br><br>NA                                                                               | 1 patient had stiffness<br>86% had normal radiographs<br>1 patient had stiffness<br>92% had normal radiographs<br><br>Significant correlation between poor results and more chronic injuries                                  |

TABLE 100.2 Results of Isolated Posterior Cruciate Ligament Reconstruction: Transtibial Versus Inlay

| Study                                                   | No. of Patients             | Age (Years) | Follow-up (Years) | Chronicity/Grade                                   | Graft Type                      | Surgical Technique        | Fixation                                       | Subjective Outcome           | Instrumented laxity                                      | Posterior Drawer                                                           | Miscellaneous                                                                                                              |
|---------------------------------------------------------|-----------------------------|-------------|-------------------|----------------------------------------------------|---------------------------------|---------------------------|------------------------------------------------|------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Kim et al. (2009), <sup>108</sup> retrospective         | Group T (transtibial): 8    | 32.4        | 3.9               | Average time to surgery, 9.4 months; all grade III | Achilles allograft in all cases | Transtibial single bundle | Tibia: bio IS<br>Femur: bio IS                 | Lysholm: 86.9                | Telos side-to-side difference: 5.6 mm                    | NA                                                                         | Significant difference in posterior translation between groups T and I2<br><br>No significant difference in Lysholm scores |
|                                                         | Group I1 (inlay): 11        | 31.9        | 11: 3.0           | all grade III                                      |                                 | AS inlay single bundle    | Tibia: bio IS<br>Femur: bio IS and washer      | Lysholm: 79.7                | 4.7 mm                                                   |                                                                            |                                                                                                                            |
|                                                         | Group I2 (inlay): 10        | 33.6        | 12: 2.5           |                                                    |                                 | AS inlay double bundle    | Tibia: bio IS<br>Femur: bio IS and washer      | Lysholm: 84.3                | 3.6 mm                                                   |                                                                            |                                                                                                                            |
| Seon and Song (2006), <sup>71</sup> retrospective       | Group A (transtibial): 21   | 29.1        | 2.6               | All chronic; all grade II or greater               | Hamstring autograft             | Transtibial               | Femur: bio IS<br>Femur: anchor screw           | Lysholm: 91.3<br>Tegner: 5.6 | Telos side-side difference: 3.7 mm                       | 19 normal/<br>grade I<br>2 grade II<br>20 normal/<br>grade I<br>2 grade II | No significant differences between groups                                                                                  |
|                                                         | Group B (tibial inlay): 22  | 29.4        | 3.0               |                                                    | Patellar tendon autograft       | Tibial inlay              | Tibia: screw and washer<br>Femur: bio IS screw | Lysholm: 92.8<br>Tegner: 6.1 | Telos side-side difference: 3.3 mm                       |                                                                            |                                                                                                                            |
| MacGillivray et al. (2006), <sup>55</sup> retrospective | Group I (transtibial): 13   | 29          | 6.3               | All chronic<br>5 grade II<br>8 grade III           | Mixed                           | Transtibial               | Tibia: IS<br>Femur: IS                         | Lysholm: 81<br>Tegner: 6     | KT posterior drawer: 5.9 mm                              | 3 grade I<br>6 grade II<br>4 grade III                                     | Neither method restored anteroposterior stability to the knee                                                              |
|                                                         | Group II (tibial inlay): 7  | 31          | 4.8               | 3 grade II<br>4 grade III                          |                                 | Tibial inlay              | Tibia: screw/washer<br>Femur: IS               | Lysholm: 76<br>Tegner: 6     | KT posterior drawer: 5.5 mm                              | 3 grade I<br>3 grade II<br>1 grade III                                     |                                                                                                                            |
| Song et al. (2014), <sup>6</sup> retrospective          | Group I (transtibial): 36   | 37          | 1                 | Unknown/All grade II or greater                    | Mixed                           | Transtibial               | Tibia: IS<br>Femur: ligament anchor screw      | Lysholm: 89.9<br>Tegner: 5.9 | KT posterior drawer: 83.3%<br>Grade I, 16.7%<br>Grade II | IKDC<br>Guidelines<br>16.7% Grade C                                        |                                                                                                                            |
|                                                         | Group II (tibial inlay): 30 | 35          | 1.1               | Unknown/All grade II or greater                    |                                 | Tibial inlay              | Tibia: two screws/washers<br>Femur: IS         | Lysholm: 92.1<br>Tegner: 6   | KT posterior drawer: 86.7%<br>Grade I, 13.3%<br>Grade II | IKDC<br>Guidelines<br>10% Grade C                                          |                                                                                                                            |

AS, Arthroscopic; bio, biologic; IKDC, International Knee Documentation Committee; IS, interference screw; KT, KT-1000 testing; NA, not available.

hamstring autograft. No significant differences were observed between the two groups with regard to functional score, ligament laxity, and radiographic changes of the knee. Hatayama and colleagues<sup>88</sup> were also unable to detect a difference in posterior tibial translation between patients treated with single- compared with double-bundle PCL reconstruction. Similarly, in a comparison of bone–patellar tendon–bone in one femoral tunnel compared with hamstring autograft in two tunnels, Houe and Jorgensen<sup>87</sup> found no difference in postoperative laxity or Lysholm and Tegner scores. Finally, a study by Deie and colleagues showed that after a mean follow-up of 12.5 years there was no significant difference in Lysholm scores, stress radiography measurements, or knee arthrometry between patients who underwent single-bundle or double-bundle PCL reconstruction with hamstring allograft.<sup>133</sup>

## COMPLICATIONS

The complications of PCL reconstruction include the more common problems that occur with orthopaedic surgery, such as infection and stiffness. In addition, some complications are associated with the specific nature of the procedure. One of the most common problems after PCL reconstruction is persistent posterior laxity. Hermans et al.<sup>112</sup> found that 11 patients in a case series of 25 patients had a positive quadriceps active test after surgery. In addition, four patients required hardware removal because of soreness, and one patient required open capsular release for postoperative arthrofibrosis and decreased ROM. The exact incidence of complication after PCL reconstruction is unknown.

Vascular injury during PCL reconstruction is rare but is a known complication. Nemani et al.<sup>113</sup> reported on a case involving a popliteal venotomy that occurred during PCL reconstruction in the setting of a previous popliteal artery bypass graft. In this case the patient had sustained a knee dislocation with vascular injury. During the PCL reconstruction, the popliteal vein was found to be adherent to the PCL remnant, and a venotomy was noted after débridement. The authors recommend caution in the setting of previous surgery because the relationship of the neurovascular structures can be altered. Although complications appear to be uncommon, the clinician should have a frank discussion with the patient preoperatively to discuss the potential risks associated with the procedure.

## FUTURE CONSIDERATIONS

The optimal treatment for the PCL-deficient knee remains unclear. Future studies will help to better define the indications for the different treatment options. These studies will attempt to overcome the limitations of prior investigations. A need exists for randomized and prospectively designed studies with controlled variables that will allow clinicians to draw definitive conclusions. The difficulties encountered in PCL study design have been discussed, but small sample size due to low incidence is probably one of the most significant. Multicenter trials will likely be required to achieve the necessary power to derive treatment recommendations. Conducting multicenter trials is difficult because of

different surgical indications and methods of reconstruction that would need to be controlled. Future efforts should focus on overcoming these limitations and thus providing data that could help elucidate the best treatment recommendations for PCL injuries.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

## SELECTED READINGS

### Citation:

Markolf KL, Zemanovic JR, McAllister DR. Cyclic loading of posterior cruciate ligament replacements fixed with tibial tunnel and tibial inlay methods. *J Bone Joint Surg Am*. 2002;84A(4):518–524.

### Level of Evidence:

Biomechanical study

### Summary:

Markolf and colleagues performed a cadaveric study comparing posterior cruciate ligament reconstruction fixed with tibial tunnel and tibial inlay techniques. Knees were subjected to 2000 cycles of tensile force of 50 to 300 N. The authors found that the inlay technique resulted in less graft failure and graft thinning.

### Citation:

McAllister DR, Petrigliano FA. Diagnosis and treatment of posterior cruciate ligament injuries. *Curr Sports Med Rep*. 2007;6(5):293–299.

### Level of Evidence:

Review

### Summary:

McAllister and Petrigliano reviewed current concepts in the diagnosis and treatment of posterior cruciate ligament injuries.

### Citation:

Yoon KH, Bae DK, Song SJ, et al. A prospective randomized study comparing arthroscopic single-bundle and double-bundle posterior cruciate ligament reconstructions preserving remnant fibers. *Am J Sports Med*. 2011;39(3):474–480.

### Level of Evidence:

II

### Summary:

Yoon and colleagues compared single- and double-bundle posterior cruciate ligament reconstruction in a prospective, randomized study. Although a small benefit was observed with regard to posterior laxity in the double-bundle group, no difference was seen with subjective outcome measures.

### Citation:

Song EK, Park HW, Ahn YS, et al. Transtibial versus tibial inlay techniques for posterior cruciate ligament reconstruction. *Am J Sports Med*. 2014;42(12):2964–2971.

### Level of Evidence:

IV, cohort study

### Summary:

Song and colleagues compare transtibial and tibial inlay PCL reconstruction techniques at an average long-term follow-up of 148 months. Clinical and radiographic outcomes were



comparable between the two methods with a significant number of patients demonstrating worsening arthritis.

**Citation:**

Fanelli GC, Edson CJ. Posterior cruciate ligament injuries in trauma patients: part II. *Arthroscopy*. 1995;11(5):526–529.

**Level of Evidence:**

IV

**Summary:**

Fanelli and Edson offer one of the few studies on the incidence of PCL injuries in trauma patients with acute hemarthrosis of the knee. More than 200 acute knee injuries with hemarthrosis were reviewed. PCL injuries occurred in 38% of acute knee injuries; 56.5% were trauma patients, and 32.9% were sports related.

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